

Laboratory Work 1

Entity–Relationship Diagram for an International Airport Database

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1. Introduction

The purpose of this laboratory work is to design an Entity–Relationship Diagram (ERD) for an International Airport database. The model describes airports, airlines, flights, passengers, bookings, boarding passes, baggage, baggage checks, security checks, and booking changes.

2. Main Entities

Entity	Primary Key	Important Foreign / Unique Keys
AIRPORT	airport_id	—
AIRLINE	airline_id	airline_code (UNIQUE)
FLIGHT	flight_id	airline_id, departure_airport_id, arrival_airport_id
PASSENGER	passenger_id	passport_number (UNIQUE)
BOOKING	booking_id	flight_id, passenger_id
BOARDING PASS	boarding_pass_id	booking_id (UNIQUE/FK)
BAGGAGE	baggage_id	booking_id
BAGGAGE CHECK	baggage_checking_id	booking_id, passenger_id
SECURITY CHECK	security_check_id	passenger_id
BOOKING CHANGE	booking_change_id	booking_id

3. Attributes and Data Types

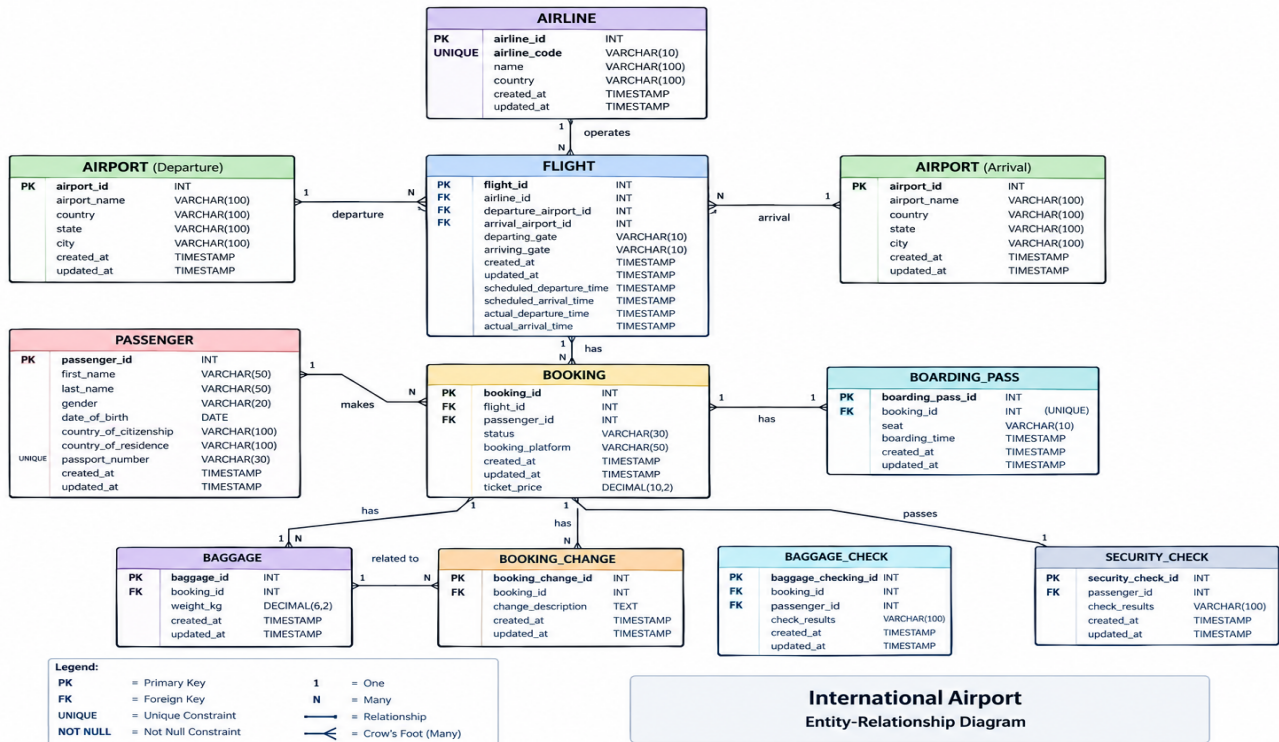
Entity	Attributes
AIRPORT	airport_id INT PK; airport_name VARCHAR(100); country VARCHAR(100); state VARCHAR(100); city VARCHAR(100); created_at
AIRLINE	airline_id INT PK; airline_code VARCHAR(10) UNIQUE; name VARCHAR(100); country VARCHAR(100); created_at TIMESTAMP
FLIGHT	flight_id INT PK; airline_id INT FK; departure_airport_id INT FK; arrival_airport_id INT FK; departing_gate VARCHAR(10); arriving
PASSENGER	passenger_id INT PK; first_name VARCHAR(50); last_name VARCHAR(50); gender VARCHAR(20); date_of_birth DATE; country
BOOKING	booking_id INT PK; flight_id INT FK; passenger_id INT FK; status VARCHAR(30); booking_platform VARCHAR(50); ticket_price
BOARDING PASS	boarding_pass_id INT PK; booking_id INT UNIQUE/FK; seat VARCHAR(10); boarding_time TIMESTAMP; created_at; updated_at
BAGGAGE	baggage_id INT PK; booking_id INT FK; weight_kg DECIMAL(6,2); created_at; updated_at
BAGGAGE CHECK	baggage_checking_id INT PK; booking_id INT FK; passenger_id INT FK; check_results VARCHAR(100); created_at; updated_at
SECURITY CHECK	security_check_id INT PK; passenger_id INT FK; check_results VARCHAR(100); created_at; updated_at
BOOKING CHANGE	booking_change_id INT PK; booking_id INT FK; change_description TEXT; created_at; updated_at

4. Relationships and Cardinality

Relationship	Cardinality
AIRLINE — FLIGHT (operates)	1 : N
AIRPORT — FLIGHT (departure)	1 : N
AIRPORT — FLIGHT (arrival)	1 : N
FLIGHT — BOOKING	1 : N
PASSENGER — BOOKING	1 : N
BOOKING — BOARDING PASS	1 : 1
BOOKING — BAGGAGE	1 : N
BOOKING — BOOKING CHANGE	1 : N
BOOKING — BAGGAGE CHECK	1 : N
PASSENGER — BAGGAGE CHECK	1 : N
PASSENGER — SECURITY CHECK	1 : N

Note: The diagram uses two visual AIRPORT boxes only to show the two roles of the same AIRPORT entity: departure airport and arrival airport. They represent one logical entity/table.

5. ER Diagram



Legend: PK = Primary Key; FK = Foreign Key; UNIQUE = Unique constraint; 1 = one; N = many. Lines show relationships between entities.

6. Textual Description

An airline operates many flights, while each flight belongs to one airline. A flight has one departure airport and one arrival airport; an airport can be used by many flights. Each flight can have many bookings, and each passenger can make many bookings. A booking is associated with a boarding pass, can have multiple baggage records, and can have multiple booking changes. Baggage checking records are associated with a booking and a passenger. Each passenger can have security check records. The model uses primary keys to uniquely identify records and foreign keys to represent relationships between tables.

7. Conclusion

The ERD provides a structured representation of the International Airport database. It defines the main entities, their attributes, key constraints, and cardinality of relationships, which can be used as a basis for implementing the relational database.